

Percutaneous Versus Surgical Cutdown Access for Transfemoral Transcatheter Aortic Valve Implantation: A Systematic Review and Meta-Analysis Focusing on Propensity-Score Matched Studies



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Background

Transcatheter aortic valve implantation (TAVI) has emerged as a potential alternative for aortic valve surgery to treat aortic valve stenosis. There is limited evidence on the comparative outcomes of TAVI access approaches, specifically the percutaneous (PC) vs surgical cutdown (SC) approach. This study aimed to assess the short-term outcomes in patients undergoing PC vs SC access for transfemoral transcatheter aortic valve replacement.

Methods

PubMed, SCOPUS, and EMBASE were searched to identify relevant studies. The primary outcomes were short-term all-cause mortality, bleeding, vascular complications, and length of in-hospital stay for patients who underwent transfemoral TAVI. Both matched and unmatched observational studies were included and subgroup analyses were performed. This systematic review and meta-analysis was performed in line with the PRISMA guidelines.

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Results

Fifteen observational studies involving 7,545 patients (3,033 underwent the PC approach and 2,466 underwent the SC approach) were included. There were no clinically significant between-group differences in short-term mortality, bleeding, length of in-hospital stay, or major vascular complications. However, minor vascular complications were significantly higher in patients who underwent PC-TAVI ($p=0.007$). In the matched subgroup, all outcomes were comparable between both groups, with the largest difference being observed in minor vascular complications more frequently occurring in the PC group ($p=0.08$).

Conclusion

The evidence shows that outcomes were comparable between the two methods of access, rendering both the PC and SC approaches equally effective for transfemoral TAVI. However, it is worth noting that minor vascular complications were more pronounced in the PC group.

Keywords

Aortic valve replacement • Transcatheter aortic valve implantation • Transcatheter aortic valve replacement • Aortic stenosis • Surgical cutdown • Percutaneous

Introduction

Cardiovascular diseases are the leading cause of death globally [1], with a significant proportion of cases being linked to valvular heart conditions. Aortic stenosis is one of the most common and serious valvular heart diseases; it occurs when the aortic valve orifice narrows and restricts blood outflow from the heart to the rest of the body, resulting in various clinical manifestations [2]. The incidence of aortic stenosis increases with age, affecting up to 10% of the population aged >70 years [3].

Transcatheter aortic valve implantation (TAVI) has emerged as a revolutionary therapeutic option for patients with severe aortic stenosis who are at a higher risk or in an inoperable state [4]. Over time, this has become an attractive alternative to surgery. The PARTNER 2 and 3 trials, the SURTAVI trial, and the Evolut 4-year follow-up trial have consistently demonstrated that TAVI is non-inferior to surgery in patients with intermediate and low surgical risk [5–8]. More than 276,000 patients underwent TAVI in the United States between 2011 and 2020 [9]. With the rising demand for TAVI procedures, there is limited evidence on the comparative outcomes of TAVI access approaches, specifically the percutaneous (PC) vs surgical cutdown (SC) approach. Both have advantages and disadvantages based on the specific patient population. This study aimed to compare transcatheter access with the surgical cutdown access technique for TAVI. By systematically evaluating clinical outcomes—including mortality, length of hospital stay, post-procedural recovery, and long-term follow-up—this study aimed to analyse which approach has more favourable benefit for patients.

Methods

Literature Search

Electronic searches were performed on Scopus, EMBASE, Cochrane, and PubMed from the database inception to June 2023 using the 'CODE' Strategy. Search terms were formulated using the Population, Intervention, Comparator, and Outcome (PICO) framework to identify studies comparing

the PC and SC approaches in adults who have undergone transfemoral TAVI. This study adhered to the Updated 2020 Version of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [10]. The keywords were as follows: (TAVI OR TAVR OR Transcatheter AND Aortic AND Valve AND Replacement OR Implantation) AND (Percutaneous OR Transcutaneous OR Transfemoral) AND (Surgical OR Cut Down OR Cut-Down OR Open Vascular). To ensure a comprehensive search and inclusion of all relevant literature, meticulous backward and forward citations were conducted. The results were initially screened through the titles and abstracts by independent authors and any conflicts were resolved by another author review. The final selection of studies was determined using the pre-established inclusion and exclusion criteria.

Inclusion and Exclusion Criteria

The main inclusion criteria were studies that compared post-operative outcomes between transfemoral PC-TAVI and SC-TAVI as an access approach. Studies that did not include at least one comparative outcome of interest were excluded from the analysis. To be included, studies had to be clinical trials or observational studies (matched and unmatched) published in the English language comparing patients who had undergone TAVI through the PC and SC approaches as the primary/initial treatment and reported either one or all of the following outcomes: (i) all-cause mortality, (ii) vascular complications (iii) bleeding, and (iv) in-hospital stay. Abstracts were included when they met the above-mentioned criteria and were the only available sources of information.

Excluded studies were systematic reviews, narrative reviews, case reports/series, editorials, study protocols, abstracts, commentaries, letters, studies that reported the outcomes for one treatment (PC/SC) without the other, and studies published in any language other than English.

Only the most recent study was included if duplicate studies were published for the same patient cohort. At both the abstract screening and full-text review stages, at least two independent authors meticulously evaluated each study and any discrepancies were resolved by another independent

reviewer. Ultimately, 15 studies including four matched studies were included in the systematic review and meta-analysis. Figure 1 illustrates the comprehensive selection process that was used to obtain the final included studies.

Definitions

The Valve Academic Research Consortium-2 (VARC-2) [11] acknowledges the fact that the Bleeding Academic Research Consortium (BARC) recently convened and established standardised bleeding definitions for patients receiving antithrombotic therapy and undergoing coronary revascularisation (percutaneous coronary intervention [PCI] or coronary artery bypass grafting). However, because the current definitions have been well adopted and shown to be accurate in predicting adverse events, the VARC-2 has chosen to maintain the original VARC definitions with BARC classifications. The VARC-2 defines life-threatening bleeding as fatal bleeding OR bleeding into a critical organ such as intracranial, intraspinal, intraocular, pericardial (requiring pericardiocentesis), or intramuscular

(resulting in compartment syndrome) OR bleeding causing hypovolaemic shock or severe hypotension requiring vaso-pressors/surgery OR bleeding with drop in haemoglobin >5 g/dL or whole blood or packed red blood cell transfusion >4 units. The VARC-2 defines major bleeding as overt bleeding associated with either a drop in haemoglobin level of at least 3 g/dL or requiring transfusion of 2 or 3 units of whole blood/red blood cells, or causing hospitalisation or permanent injury or requiring surgery. The VARC-2 defines minor bleeding as any bleeding worthy of clinical mention (eg, access site haematoma) that does not qualify as life-threatening, disabling, or major.

The VARC-2 classifies vascular complications as major and minor. Major vascular complications include: any aortic dissection, aortic rupture, annulus rupture, left ventricle perforation, or new apical aneurysm/pseudoaneurysm OR access site or access-related vascular injury (dissection, stenosis, perforation, rupture, arterio-venous fistula, pseudoaneurysm, haematoma, irreversible nerve injury, compartment syndrome, percutaneous closure device

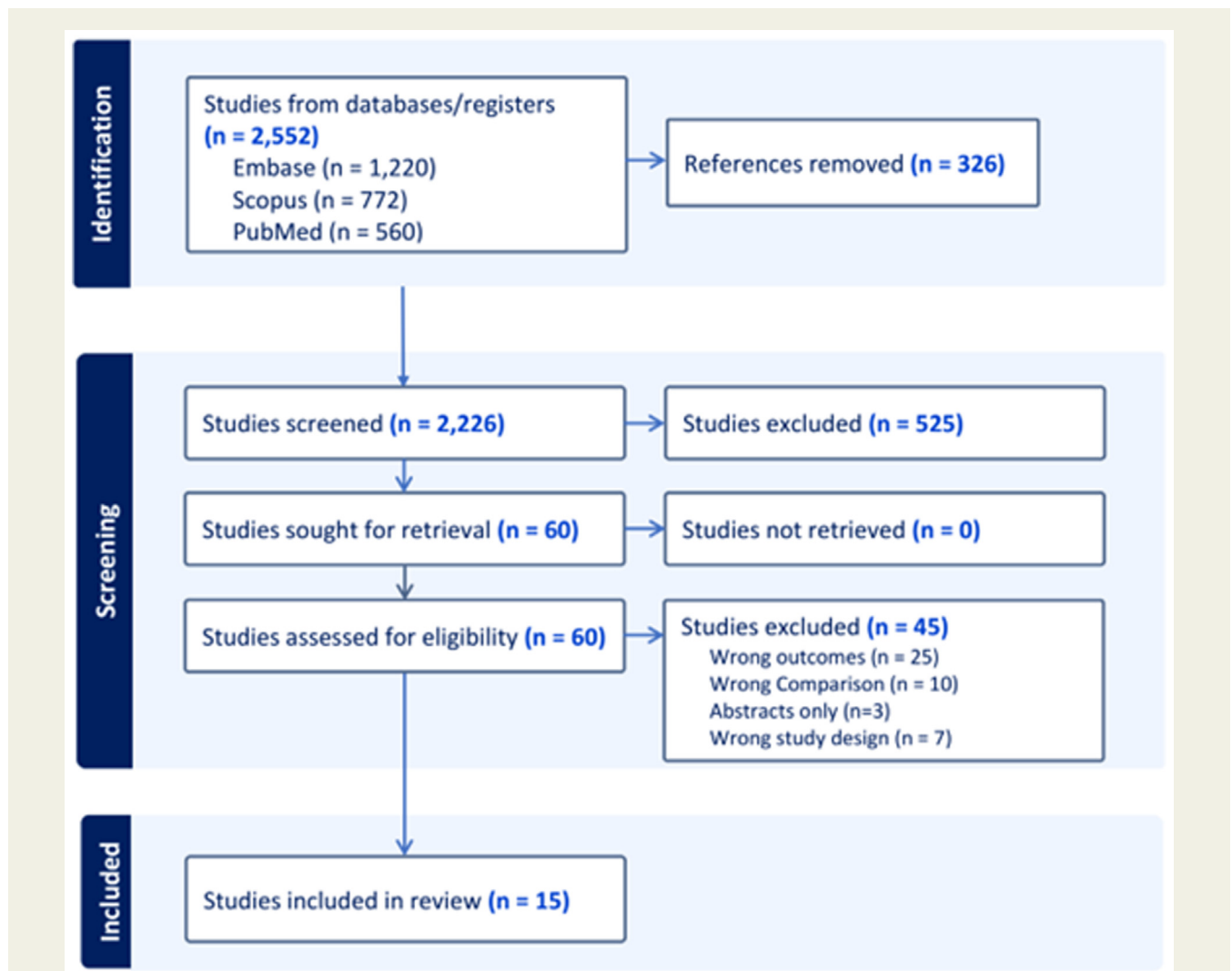


Figure 1 PRISMA flow diagram illustrating the course through which the studies were selected.

failure) leading to death, life-threatening or major complications. Minor vascular complications include access site or access-related vascular injury (dissection, stenosis, perforation, rupture, arterio-venous fistula, pseudoaneurysms, haematomas, percutaneous closure device failure) not leading to death, life-threatening or major complications.

Data Extraction and Quality Assessment

Two independent investigators were responsible for data extraction from each included article. The obtained data were further revised by the other two members, and any conflicts were resolved. Consistency and accuracy were therefore ensured.

The extracted data focused on study characteristics, key demographic data, including age, sex, body mass index (BMI), and the occurrence of comorbidities such as cerebrovascular accident, peripheral vascular disease, hypertension, and diabetes mellitus. Extraction also focused on previous cardiac interventions, such as prior PCI and atrial fibrillation. Aortic valve characteristics—such as aortic valve area, valve size, sheath size, balloon post-dilation, and valve-in-valve procedures—were also considered. The mode of anaesthesia, platelet count, and international normalised ratio were also included in the patient characteristics.

Categorical data were extracted as events and the total for each group, while continuous data were coded as mean and standard deviation. If the data were reported in other formats, the method described by Wan *et al.* [12] was used to perform the necessary conversion.

The main outcomes included short-term (in-hospital or 30-day) all-cause mortality, in-hospital length of stay, vascular complications (major and minor), and bleeding events (major and minor). The secondary outcomes included life-threatening bleeding complications, minor vascular complications, minor bleeding events, and acute renal failure. All outcomes were defined according to VARC-2 criteria (Definitions section).

A quality appraisal schema based on the Newcastle–Ottawa scale risk of bias assessment tool was used to assess all included studies [13]. Two reviewers independently assessed the risk of bias, and a final table was assembled based on their agreement (Table 1). The evaluation cutoff for the follow-up length was set at 30 days. Follow up was considered sufficient if no more than 10% of the patient cohort data were lost.

Statistical Analysis

This meta-analysis adhered to the recommendations provided by the Cochrane Collaboration and Meta-analysis of Observational Studies in Epidemiology (MOOSE) [14]. Review Manager software version 5.4.1 (Cochrane Collaboration) were used for data analysis. The Mantel–Haenszel random-effects model was employed to estimate the risk ratio (RR) and the corresponding 95% confidence intervals (CIs) for binary outcomes. The weighted mean difference for the continuous outcomes was calculated using the inverse variance method. To assess statistical heterogeneity, the

Q-test was used for heterogeneity (Cochrane, 1954) and I^2 statistics, where an I^2 value $>50\%$ was considered an indicator of substantial heterogeneity. A p -value <0.05 was considered a statistically significant outcome. Sensitivity analysis was conducted to test the robustness of the results.

Results

Demographic Characteristics

Fifteen studies [15–29] involving 7,545 patients (3,033 underwent the PC approach and 2,466 underwent the SC approach) were included in this meta-analysis for the transfemoral TAVI access group. Analysis of the demographic profiles of patients who underwent PC and SC access revealed no significant differences in terms of age and BMI. A detailed summary of baseline demographic characteristics is provided in Table 2.

Overall, the patients who underwent PC-TAVI had a comparable risk of all-cause mortality to SC-TAVI (RR 1.08, 95% CI 0.76–1.54, $p=0.68$, $I^2=28\%$). Similarly, in the matched subgroup, mortality rates after PC-TAVI and SC-TAVI were similar (RR 1.06, 95% CI 0.63–1.77, $p=0.82$, $I^2=25\%$). The overall risk of all-cause mortality was slightly higher with the PC approach than with the SC approach; however, these differences were not statistically significant (Figure 2). The analysis of the TAVI access approach revealed that life-threatening bleeding was slightly higher in the PC approach (RR 1.23; 95% CI 0.77–1.98, $p=0.39$; $I^2=0\%$); however, the results were not statistically significant (Figure 3).

Overall, the risk of major bleeding was higher with the PC approach than with the SC approach (RR 1.28; 95% CI 0.84–1.95, $p=0.24$; $I^2=51\%$); however, the differences were not statistically significant (Figure 4). The results were also similar in the matched subgroup.

The pooled risk of minor bleeding was higher in the PC-TAVI group when compared with SC-TAVI (RR 1.21, 95% CI 0.74–1.97, $p=0.46$, $I^2=80\%$); however, these differences were not statistically significant. By contrast, the PC approach had a lower risk of minor bleeding (RR 0.71; 95% CI 0.45–1.10, $p=0.13$; $I^2=0\%$) in the matched subgroup, with no statistically significant result when compared with the SC approach.

The incidence of major vascular complications was similar in both groups (RR 1.10, 95% CI 0.87–1.39, $p=0.43$, $I^2=4\%$), and no statistically significant difference was noted (Figure 5).

The overall incidence of minor vascular complications was higher in the PC-TAVI than in the SC-TAVI groups, with statistically significant results (RR 1.91, 95% CI 1.20–3.05, $p=0.007$, $I^2=74\%$) (Figure 6). In the matched group, PC-TAVI had a higher risk of minor vascular complications; however, the results were not statistically significant (RR 1.75, 95% CI 0.93–3.28, $p=0.08$, $I^2=73\%$) when compared with the PC approach.

In-hospital stay was comparatively lower in the PC approach (MD -0.55, 95% CI -1.65–0.55, $p=0.33$, $I^2=90\%$), but the results were not statistically significant. Similar results were noted in the matched subgroup (MD -0.59, 95% CI -3.18–1.99, $p=0.65$, $I^2=94\%$).

Table 1 Quality assessment using the Newcastle–Ottawa scale (follow-up length was accepted as in-hospital stay and adequacy of follow-up meant <10% loss).

Study ID	Selection				Comparability		Outcome			Score
	Representativeness of the exposed	Selection of non-exposed	Ascertainment of exposure	Outcome of interest not present at start of study	Main factor	Additional factor	Assessment	Follow-up length	Adequacy of follow-up	
Arifi et al. 2022 [15]	★	★	★	★	★	0	★	★	★	8
Drafts et al. 2017 [16]	★	★	★	★	★	0	★	★	★	8
Eckner et al. 2021 [17]	★	★	★	★	★	0	★	★	★	8
Ceresa et al. 2020 [18]	★	★	★	★	★	★	★	★	★	9
Bernardi et al. 2015 [19]	★	★	★	★	★	0	★	★	★	8
Kochman et al. 2018 [20]	★	★	★	★	★	★	★	★	★	9
Stathogiannis et al. 2021 [21]	★	★	★	★	★	0	★	0	0	6
Nakamura et al. 2014 [22]	★	★	★	★	★	0	★	★	★	8
Gennari et al. 2021 [23]	★	★	★	★	★	★	★	★	★	9
Enriquez et al. 2016 [24]	★	★	★	★	★	0	★	★	★	8
Lebehn et al. 2016 [25]	★	★	★	★	★	0	★	0	0	6
Mach et al. 2018 [26]	★	★	★	★	★	0	★	★	★	8
Kadakia et al. 2014 [27]	★	★	★	★	★	★	★	★	★	9
Kedev et al. 2019 [28]	★	★	★	★	★	0	★	★	★	8
Spitzer et al. 2015 [29]	★	★	★	★	★	0	★	★	★	8

Table 2 Summary of meta-analysis outcomes using the Newcastle–Ottawa scale (follow-up length was accepted as in-hospital stay, adequacy of follow-up meant <10% loss).

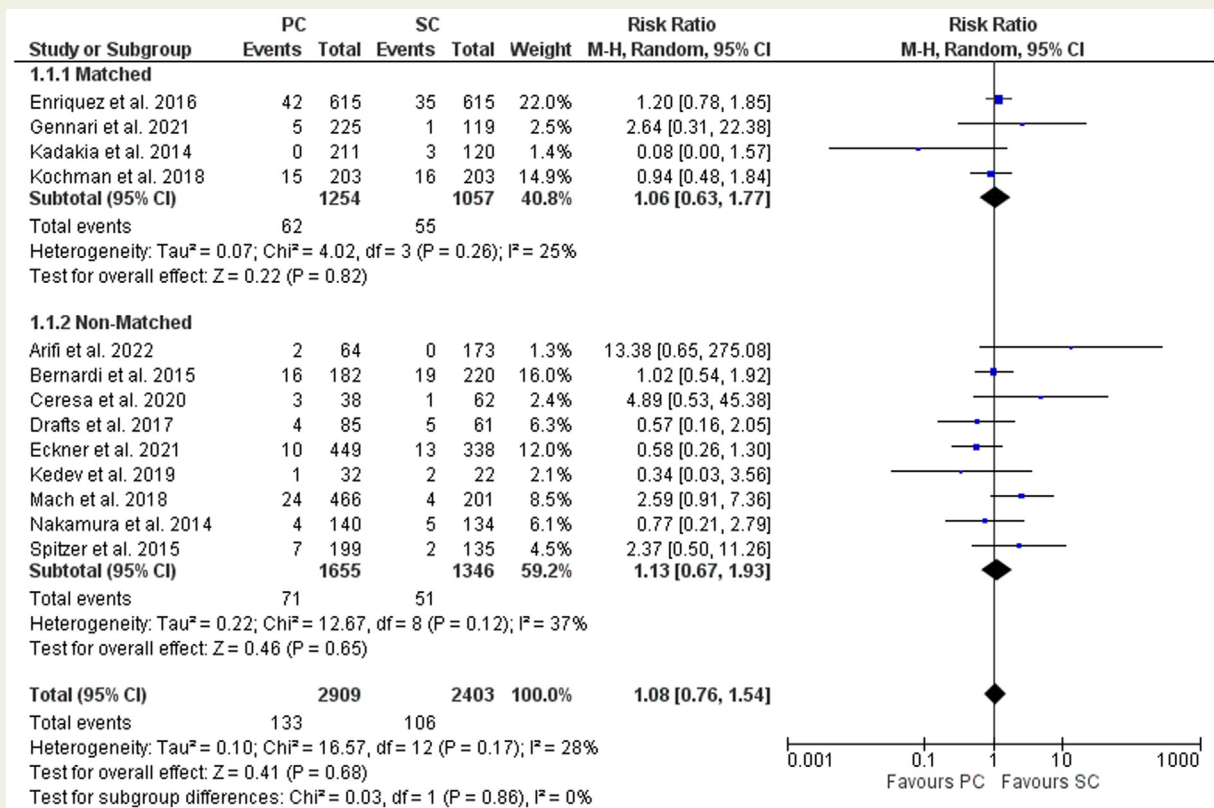
Study ID	No of subjects		Age (years)		Male		BMI		HTN		DM	
	PC	SC	PC	SC	PC	SC	PC	SC	PC	SC	PC	SC
Arifi et al. 2022 [15]	64	173	72 (68–77)*	74 (70–79)*	33	80	28.9 (24.8–33.6)*	35.2 (30.2–39.7)*	45	109	34	107
Drafts et al. 2017 [16]	85	61	77.6±9.9	82.6±0.4	48	38	28.6±7.7	28±6.2	-	-	-	-
Eckner et al. 2021 [17]	449	338	81±5.6	81.4±6.1	220	150	27.3±5.1	27.1±4.6	-	-	127	122
Ceresa et al. 2020 [18]	38	62	83.1±5.12	84.9±5.5	-	-	-	-	-	-	-	-
Bernardi et al. 2015 [19]	182	220	81.6±7.3	81.5±8	83	108	26.3±4.7	25.9±4.3	-	-	56	71
Kochman et al. 2018 [20]	203	203	79.1±7.2	78.8±8.5	95	86	27.4±4.3	28±4.9	-	-	-	-
Nakamura et al. 2014 [22]	140	130	84±7.5	84.5±7.4	81	79	27±6	26.3±5.2	123	128	57	54
Gennari et al. 2021 [23]	225	119	83 (79–86)*	83 (78–85)*	130	67	25.1 (23.4–28)*	24.17 (22.3–27.5)*	180	89	52	27
Enriquez et al. 2016 [24]	615	615	82±7	81±7	272	281	28±5	28±5	503	494	226	224
Mach et al. 2018 [26]	466	201	81.5±5.2	80.5±4.7	159	82	27±4.8	27.4±4.7	427	194	159	76
Kadakia et al. 2014 [27]	211	120	85 (80–89)*	85 (81–89)*	102	65	29.3±13.3	28.1±6.9	188	107	67	44
Kedev et al. 2019 [28]	32	22	76±6.9	73.6±6.8	17	9	28±3.9	27.6±4.2	30	21	9	6
Spitzer et al. 2015 [29]	199	135	81.4±4.6	81.1±4.4	65	55	27.2±4.6	27.8±4.3	195	135	84	56

Abbreviations: PC, percutaneous; SC, surgical cut down; BMI, body mass index; HTN, hypertension; DM, diabetes mellitus. Continuous variables are represented by mean±standard deviation except stars (*) represented by median and interquartile ranges.

Discussion

Transcatheter aortic valve implantation (TAVI) has gained popularity since the first percutaneous aortic valve

implantation in 2002. Patients who undergo TAVI are often elderly adults with concomitant illnesses that impede surgical aortic valve replacement (SAVR) [30,31]. After the PARTNER trials demonstrated the non-inferiority of TAVI compared with

**Figure 2** Forest plot for all-cause mortality.

Abbreviations: PC, percutaneous; SC, surgical cut-down; M-H, Mantel–Haenszel; RE, random-effects model; RR, risk ratio; CI, confidence interval.

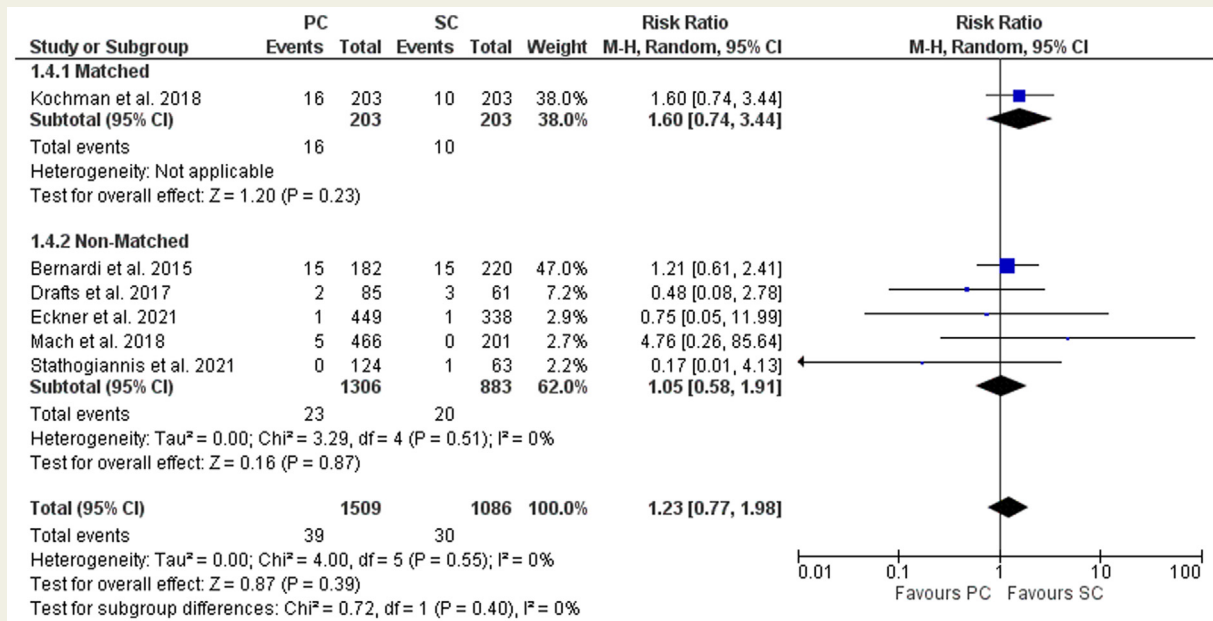


Figure 3 Forest plot for life-threatening bleeding.

Abbreviations: PC, percutaneous; SC, surgical cut-down; M-H, Mantel-Haenszel; RE, random-effects model; RR, risk ratio; CI, confidence interval.

the conventional surgical approach in terms of in-patient mortality for patients with severe aortic stenosis up to 3 years, transcatheter aortic valve replacement (TAVR) began to be recognised as an efficient alternative treatment modality [5–7].

Clinical practice guideline committees have examined increasing amounts of data over the past 10 years and identified TAVI as a viable substitute for SAVR in patients at high surgical risk [30,31]. As TAVI has emerged as an

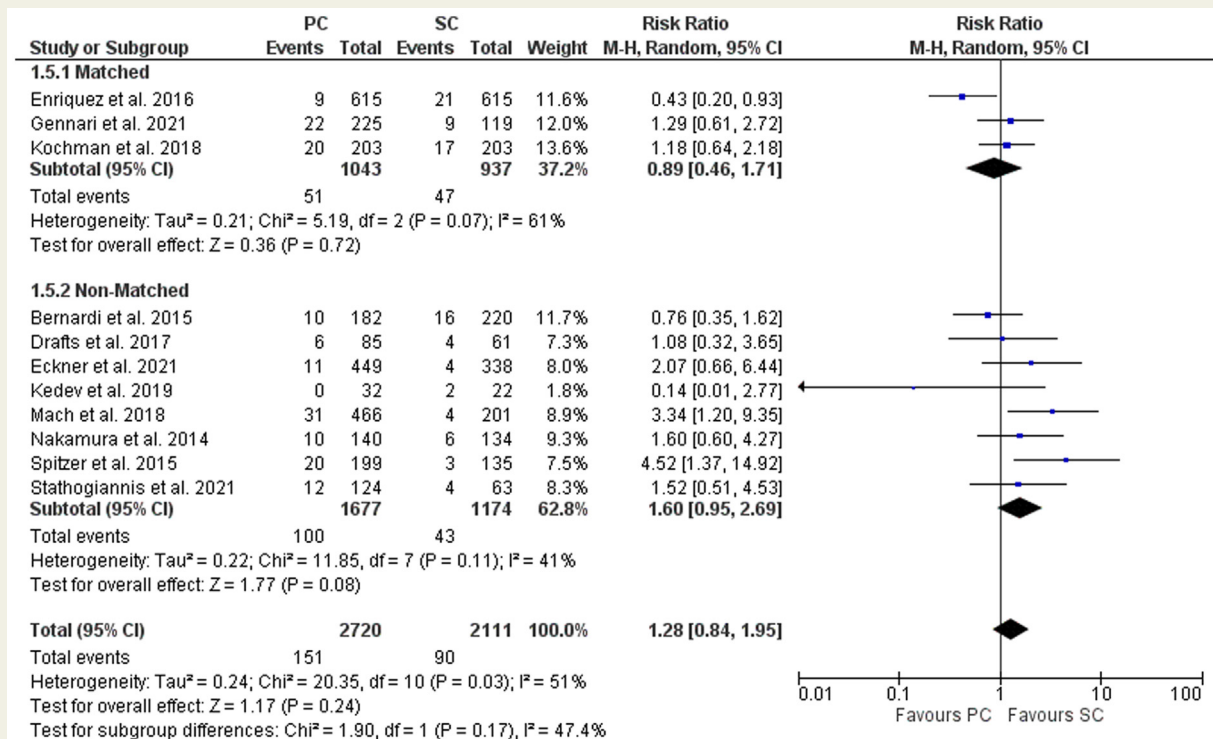


Figure 4 Forest plot for major bleeding.

Abbreviations: PC, percutaneous; SC, surgical cut-down; M-H, Mantel-Haenszel; RE, random-effects model; RR, risk ratio; CI, confidence interval.

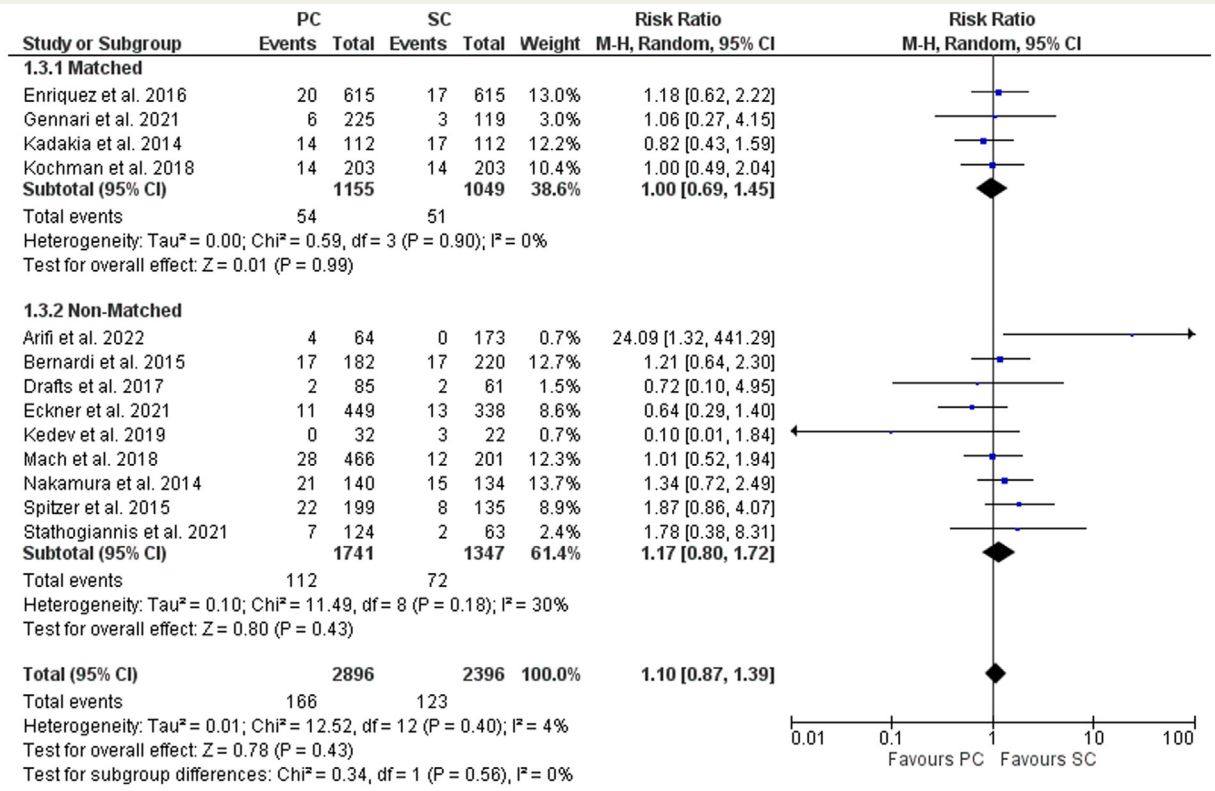


Figure 5 Forest plot for major vascular complications. Abbreviations: PC, percutaneous; SC, surgical cut-down; M-H, Mantel-Haenszel; RE, random-effects model; RR, risk ratio; CI, confidence interval.

efficient replacement for SAVR, a crucial clinical query has arisen: out of PC and SC femoral access approaches for TAVI, which method has more favourable outcomes? This review aimed to answer this question by compiling results from studies that compared PC with SC access in TAVI. The findings showed that patients who underwent TAVI using the PC approach had comparable results with those who underwent SC-TAVI. Mortality rates were numerically, but not significantly, higher in the PC-TAVI group. The preference for the SC-TAVI seemed to be more prevalent among patients who were obese and had complex anatomical considerations. Nevertheless, there is still no definitive evidence regarding the superiority of either approach.

Recently published studies have concluded that the SC and PC approaches for transfemoral TAVR are equally safe. This review found a noticeable trend indicating higher incidences of bleeding and major and minor vascular complications within the PC-TAVI group; however, the results were not statistically significant. These complications were assessed using the Vascular Events Classification System for Transcatheter Aortic Valve Repair and Replacement (VERC-2) criteria [11].

Analysis of the demographic data from the included studies indicates that certain factors—including the sheath-to-femoral artery ratio, sheath diameter, female gender,

pre-existing peripheral arterial disease, and dyslipidaemia—hold potential as significant predictive indicators for major vascular complications in patients undergoing TAVI procedures [32,33].

A noteworthy outcome was observed in terms of minor vascular complications within the unmatched TAVI-SC subgroup ($p=0.04$), with a similar trend maintaining near significance in the matched subgroup ($p=0.08$). The elevated prevalence of vascular complications in the PC-TAVI group could potentially be attributed to certain anatomical characteristics of the vessel, including diameter, degree, and distribution of calcification, as well as the tortuosity of the vessel. Additionally, the surgical expertise of the vascular surgeon may play a role. Importantly, this phenomenon is believed to be connected to the more direct and controlled access to the femoral artery during SC-TAVI. It is important to note that percutaneous puncture involves a relatively less precise entry, which could contribute to the higher risk associated with PC-TAVI access compared with SC access [22,33].

The meta-analysis showed a slightly higher overall risk of bleeding in the PC-TAVI group. Both major and minor bleeding issues were more prevalent with the PC approach. However, no statistically significant differences were discerned in either the SC- or PC-matched subgroups.

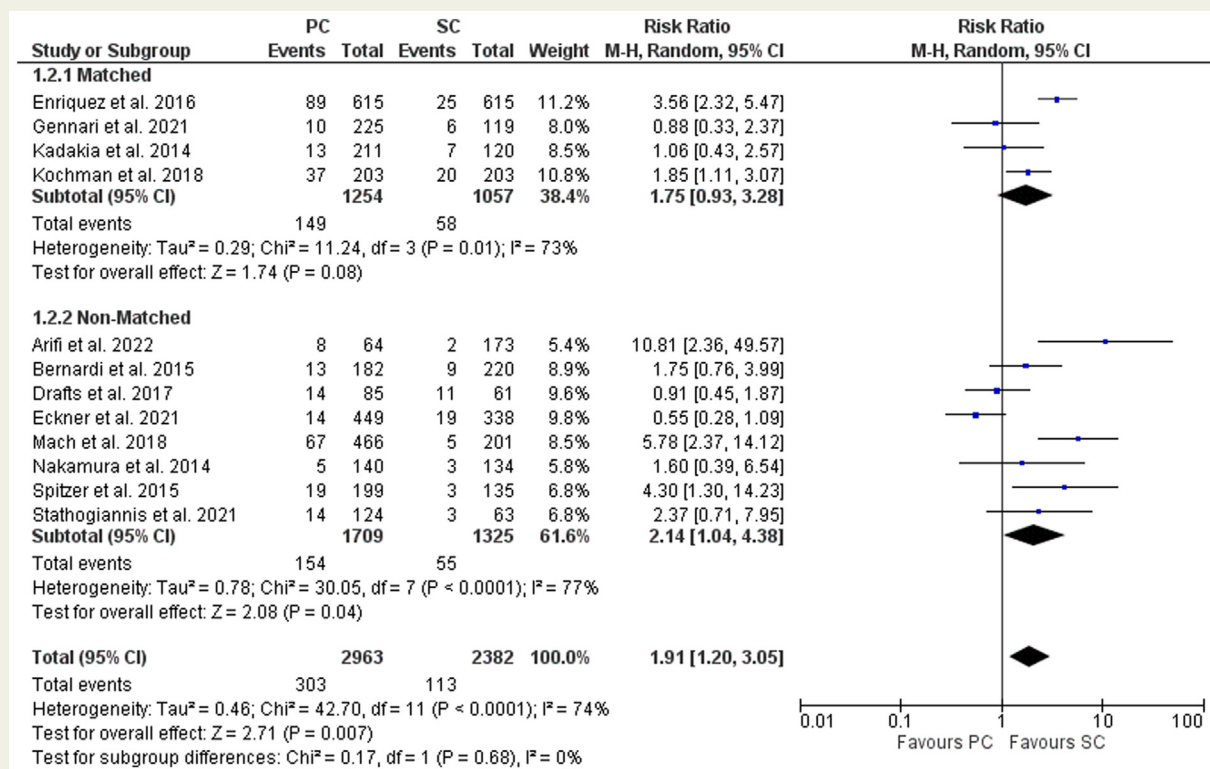


Figure 6 Forest plot for minor vascular complications.

Abbreviations: PC, percutaneous; SC, surgical cut-down; M-H, Mantel-Haenszel; RE, random-effects model; RR, risk ratio; CI, confidence interval.

Vascular bleeding incidents could be linked to the sheath size used, as larger sheath sizes are associated with elevated bleeding risk. Kockman et al. showed that using smaller sheath sizes (18 French) resulted in significantly fewer major bleeding complications [20]. However, these results oppose those of Mach et al. [27], who demonstrated that the SC group exhibited significantly fewer bleeding issues ($p=0.01$) compared with the PC group, where minor vascular complications were also notably common in the PC group ($p=0.04$). Nonetheless, there is a higher possibility of wound-related complications, such as wound infection or seroma, following SC-TAVI procedures. However, the reporting of these complications was less consistent in the literature.

The pooled analysis demonstrated a trend towards shorter length of hospital stay with the PC group, so they are frequently discharged home earlier, thus reducing procedural costs. It is important to note the significant heterogeneity was observed in this outcome across the included studies. The slower recovery of the SC group may put them at risk of prolonged immobilisation and higher infection rates [16,21]. According to Kadokia et al. [28], the shorter hospital stay in the PC group may have been due to quicker recovery and time to ambulation. In contrast to the findings by Eckner et al., the PC strategy was unexpectedly linked to a longer hospital stay [18].

While most outcomes demonstrate similarity between both approaches, it is imperative to adopt a patient allocation strategy that considers their baseline characteristics and comorbidities, with specific emphasis on anatomical anomalies and obesity. Noteworthy, there has been a discernible shift in the type of anaesthesia used in transfemoral TAVI procedures, with markedly lower complications with local anaesthesia than general anaesthesia [34]. Consequently, patients who are suited for local anaesthesia are ideally guided towards the PC approach, as SC procedures often necessitate general anaesthesia, potentially leading to increased complications.

Limitations and Future Directions

This systematic review and meta-analysis had several limitations. The analysis exclusively focused on propensity-matched and retrospective cohort observational studies, introducing inherent bias. Although this approach was chosen due to the absence of randomised controlled clinical trials in the literature, it limited the study design. Furthermore, the use of different types of valve implantation devices across the included studies introduced potential variability in outcomes due to differences in design, durability, and repositioning capabilities. The varying sample sizes among the studies could also have influenced the research findings. Additionally, reliance on data from national registries in

some studies introduced the possibility of subject bias. Selection bias may have also played a role, as the choice of access site strategy was influenced by individual and centre-specific preferences. Patients with obesity or complex anatomy are frequently directed towards the surgical route, leading to bias in patient selection. It is essential to recognise that the devices used in these trials were deployed over an extended period, allowing for the emergence of newer generations; how this relates to outcomes was not investigated. Future studies are encouraged to explore the differences observed with newer generations of devices. It is evident that there is a need for new prospective studies and randomised controlled clinical trials to validate the study's findings and comprehensively address these limitations.

Conclusion

This analysis of the current evidence indicates that both access site strategies exhibit comparability in terms of all-cause mortality, major vascular complications, and bleeding complications. However, it is worth noting that minor vascular complications were more pronounced in the PC group.

Funding

No funding was received to perform this study.

Conflict of Interest

There is no conflict of interest to declare.

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